

$$\frac{2}{3}$$

$$\frac{2}{3}$$

$$\frac{2}{3}$$

$$-\frac{2}{3}$$

$$0$$

$$\frac{5}{6}$$

$$\frac{1}{4}$$

$$\frac{1}{2}$$

$$\frac{5}{6}$$

$$\frac{5}{7}$$

$$0$$

$$\infty$$

$$\infty$$

$$\infty$$

$$\frac{8}{27}$$

$$\frac{9}{4}$$

$$-\frac{8}{27}$$

$$\frac{2}{3}$$

$$1$$

$$\frac{2}{3}$$

$$\frac{2}{3}$$

$$-2$$

$$\frac{13717421}{219478738}$$

$$1267650600228229401496703205376$$

$$16$$

$$2y^3x^2|_{x=1,y=2}=16$$

$$16$$

$$2y^3x^2$$

$$\frac{d}{dx}2y^3x^2=4y^3x$$

$$4y^3x$$

$$\frac{d}{dy}2y^3x^2=6y^2x^2$$

$$6y^2x^2$$

$$\frac{d}{dz}2y^3x^2=0$$

$$0$$

$$\int_1^2 2y^3x^2dx=\frac{14y^3}{3}$$

$$\frac{14y^3}{3}$$

$$\int_1^2 2y^3x^2dy=\frac{15x^2}{2}$$

$$\frac{15x^2}{2}$$

$$2$$

$$\int_1^2 2y^3x^2dz = 2y^3x^2$$

$$2y^3x^2$$

$$\frac{d}{dz}2y^3x^2=0$$

$$\frac{d}{dy}2y^3x^2=6y^2x^2$$

$$\frac{d}{dx}2y^3x^2=4y^3x$$

$$2y^3x^2+4y^3x+6y^2x^2$$

$$\frac{d}{dz}2y^3x^2=0$$

$$\frac{d}{dy}2y^3x^2=6y^2x^2$$

$$\frac{d}{dx}2y^3x^2=4y^3x$$

$$2y^3x^2+4y^3x\geq 6x^2y^2$$

$$\frac{d}{dz}2y^3x^2=0$$

$$\frac{d}{dy}2y^3x^2=6y^2x^2$$

$$\frac{d}{dx}2y^3x^2=4y^3x$$

$$\frac{d}{dx}2y^3x^2+4y^3x\geq 6x^2y^2=4y^3+4y^3x\geq 12y^2x$$

$$4y^3+4y^3x\geq 12y^2x$$

$$\frac{5x}{7z}+\frac{4y}{7z}$$

$$2$$

$$2y$$

$$y+z$$

$$3$$

$$x$$

$$y+2z$$

$$\frac{x^2-y^2}{x-y}$$

$$\frac{x^2+y^2+2xy}{x+y}$$

$$x+z$$

$$2y+3z$$

$$3xy$$

$$\frac{x+y}{x+z}$$

$$y+\frac{z}{x}$$

$$1+\frac{y}{x}$$

$$2x+3y$$

$$1$$

$$0$$

$$\int_2^4 x^2+zd{x}=2z+\frac{56}{3}$$

$$2z+\frac{56}{3}$$

$$f(x,y)=\left\{\begin{array}{ll}\frac{2x^{\frac{27}{10}}y^{\frac{13}{5}}+11}{2y^{\frac{13}{5}}}&x\geq y\\ \frac{2y^{\frac{27}{10}}x^{\frac{13}{5}}+11}{2x^{\frac{13}{5}}}&x< y\\ 0&otherwise\end{array}\right.$$